Autoliv AUTOLIV (CHINA) ELECTRONICS	SHA Brake System EOL Test Specification	Date: 2014.10.08 Rev.: D

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SHA Brake System EOL Test Specification
SHA 制动系统下线检测规范

SUBJECT: This document defines the process flow for SHA brake system EOL test and all related diagnostic communication services.
这份文件描述了 SHA 制动系统下线检测流程以及相关联通讯信号的详细信息。

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A	Kailai Sun	2014.08.26	Initial Release
B	Kailai Sun	2014.09.11	Delete security access service, change description in Section 5.4, and update some description
C	Kailai Sun	2014.09.30	Update Dynamic test data flow, and service 2F content
D	Kailai Sun	2014.10.08	Update service 2F request and response format

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1 Reference Documents 参考文档

The following is a list of documents used to establish this document.

Nº	Document	Reference	Source
1			
2			

2 Terminology 专用术语

The following is a list of terms and acronyms that are in the document.
以下为本文档涉及的相关专用术语

IRBC - Integrated Restraints and Brake Controller 约束和制动整合控制器

SHA - Smart Hydraulic Actuator 智能液压控制器

DTC – Diagnostic Trouble Code 故障诊断代码

EOL – End of Line 下线检测

OBD – On Board Diagnostics 车载诊断接口

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3 Introduction 介绍

The purpose of EOL test is to verify the IRBC-SHA system is correct installed and function properly on the vehicle. This document defines the process flow for EOL test and all related communication messages.

下线检测的目的是验证 IRBC-SHA 系统的正确安装和正常工作。本文档定义了针对 IRBC-SHA 系统的下线检测流程和相关通讯信号。

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4 EOL Test Procedure 下线检测流程

The EOL test consists of 15 steps. Overall EOL test procedure is list in Table 1.

整个下线检测流程一共有 15 个步骤，具体内容参见表 1。

Table 1: EOL Test Procedure

Step	Description	Diagnostic Mode	Message Description	Hex message
1	Start of Test preparation, drive vehicle into test bay, connect Test Tool to vehicle	Default	Default Diagnostic Session Request	02 10 01 00 00 00 00 00
2	Verify IRBC and SHA part numbers to ensure correct installation.	Default	Read IRBC Part Number Read SHA Part Number	03 22 F1 CB 00 00 00 00 03 22 F1 13 00 00 00 00
3	Read and store DTCs	Default	Read DTC	03 19 02 8B 00 00 00 00
4	Clear stored DTCs	Default	Clear DTC	04 14 FF FF FF 00 00 00
5	Verify Brake Switch Input	Default	Verify Vehicle CAN for Brake Switch	
6	Verify Wheel Speed Sensor Output for vehicle at rest	Default	Verify Vehicle CAN for Wheel Speeds	
7	Verify operation of Warning Lamps	Extended	Extended Diagnostic Session Tester Present (Periodic @ 2s rate) Lamp Control All On Lamp Control All Off	02 10 03 00 00 00 00 00 02 3E 00 00 00 00 00 00 07 2F 32 00 03 7E 00 00 04 2F 32 00 00 00 00 00
8	End of Static Test / Verify NO DTCs stored	Default	Read DTC	03 19 02 8B 00 00 00 00
9	Start of Dynamic Test	Extended	Extended Diagnostic Session Tester Present (Periodic @ 2s rate)	02 10 03 00 00 00 00 00 02 3E 00 00 00 00 00 00
10	Verify Wheel Speed Sensor Input signal is correct	Extended	Verify Vehicle CAN for Wheel Speeds Tester Present (Periodic @ 2s rate)	02 3E 00 00 00 00 00 00
11	Verify operation of base brake system	Extended	Tester Present (Periodic @ 2s rate)	02 3E 00 00 00 00 00 00

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12	Verify operation & correlation of pressure build, confirm brake pipe routing and proper wheel speed sensor wiring connections	Extended	Tester Present (Periodic @ 2s rate) FL pressure build FL pressure release FR pressure build FR pressure release RL pressure build RL pressure release RR pressure build RR pressure release Motor only All off	02 3E 00 00 00 00 00 00 07 2F 31 07 03 40 53 00 07 2F 31 07 03 43 53 00 07 2F 31 07 03 10 A3 00 07 2F 31 07 03 1C A3 00 07 2F 31 07 03 04 A3 00 07 2F 31 07 03 34 A3 00 07 2F 31 07 03 01 53 00 07 2F 31 07 03 C1 53 00 07 2F 31 07 03 00 03 00 04 2F 31 07 00 00 00 00
13	Perform Vacuum Relief	Extended	Tester Present (Periodic @ 2s rate) Open Prime Valves for 2 seconds	02 3E 00 00 00 00 00 00 07 2F 31 07 03 00 C1 00 (after 2 seconds) 04 2F 31 07 00 00 00 00
14	Verify NO DTCs are stored	Extended	Read DTC Tester Present (Periodic @ 2s rate)	03 19 02 8B 00 00 00 00 02 3E 00 00 00 00 00 00
15	**End** of the 'End-of-Line' test sequence	Extended	Clear DTC Reset	04 14 FF FF FF 00 00 00 02 11 01 00 00 00 00 00

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4.1 Step 1: Start of EOL Test

The operator starts the vehicle and the vehicle will complete an initialization check. If no functional warning lamps are displayed on the cluster, the operator drivers the vehicle onto the EOL test facility and then establish communication between the vehicle and the process equipment via OBD. The process equipment sends a Default Diagnostic Session request to the IRBC and a Positive Response shall be sent from the IRBC to the process equipment if communication can be established properly. Refer to Section 5.1 for detailed communication messages.

操作员启动车辆，待车辆完成自检。如果没有发现仪表盘有报警灯亮起，则将车辆驶入测试台，然后将车辆与测试设备通过 OBD 端口连接，并建立通讯。测试设备向 IRBC 发送 Default Diagnostic Session 报文，如果通讯正常，IRBC 会反馈给测试设备 Positive Response 报文。第 5.1 节中对相关的通讯报文作了详细定义。

4.2 Step 2: Verify IRBC and SHA Part Numbers

Part numbers for IRBC and SHA will be verified to ensure IRBC and SHA installations match with the vehicle configuration. Refer to Section 5.7 and 5.8 on detailed messages for reading IRBC and SHA part numbers.

验证 IRBC 和 SHA 模块的零件号以确保这两个模块和车辆的配置相匹配。第 5.7 和 5.8 节详细定义了相关的通讯报文。

4.3 Step 3: Read and Store DTCs

Refer to Section 5.6 for reading DTCs. The Process Equipment will need to store what is read from the IRBC.

读取 DTC 码的通讯报文可以参见第 5.6 节。检测设备需记录所读取到的 IRBC DTC 码内容。

4.4 Step 4: Clear DTCs

During the vehicle assembly, various DTCs could be set and stored in IRBC if the IRBC is powered up with some modules missing. These DTCs need to be cleared during EOL test. Refer to Section 5.5 for clearing stored DTCs.

装车过程中，在所有模块没有安装到位的情况下给 IRBC 上电会报错，生成的 DTC 码会存储在 IRBC 中。在下线检测时这些 DTC 码需要被清除。相关的通讯报文定义可参见第 5.5 节。

4.5 Step 5: Verify Brake Switch Input

Apply / release the brake and ensure that the process equipment indicates the correct brake pedal status. Refer to CAN diag spec for detailed brake pedal status in service $22(DID $2D01).

通过踩/松制动踏板来确认测试仪器是否能够正常反映制动踏板状态。有关制动踏板状态信息在诊断规范里的服务$22(DID $2D01)中有作详细定义。

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4.6 Step 6: Verify Wheel Speed Sensor Input while at Rest

The process equipment shall indicate readings for all the wheel speed sensors are less than 1 km/h. Refer to CAN diag spec for detailed wheel speed information in service $22(DID $2D00).

在车辆静止的状态下，测试设备读取到的轮速传感器值应小于 1 km/h。有关轮速传感器信息在诊断规范里的服务$22(DID $2D00)中有作详细定义。

4.7 Step 7: Verify Operation of Warning Lamps

The process equipment sends out three signals in sequence (as defined in Figure 1) to the IRBC to start Extended Diagnostic Service (defined in Section 5.1). The process equipment then sends out InputOutputControlbyIdentifier messages (defined in Section 5.4) to control the On/Off state of related warning lamps (defined in Figure 2) on the cluster. This process is to ensure each warning lamp operates properly.

测试设备依次向 IRBC 发送三个信号（见图 1）来开启 Extended Diagnostic Service（见第 5.1 节定义）。然后测试设备通过向 IRBC 发送 InputOutputControlbyIdentifier 报文来控制仪表盘上相关报警灯（见图 2）的亮/灭。这道工序意在确认报警灯的正常工作。

Figure 1: Signal Sequence

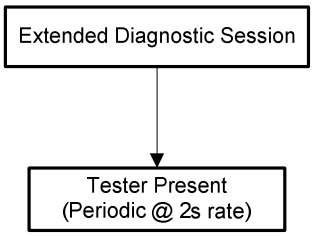

Figure 2: List of Warning Lamps

Brake Warning Lamp	
ABS Warning Lamp	
Airbag Warning Lamp	
Seat Belt Lamp	
ESC Active / Fault Lamp	

4.8 Step 8: End of Static Test

Read DTCs stored in the IRBC to ensure no DTCs was set and stored in the IRBC from Step 1 to Step 7.

通过读取 IRBC 中的 DTC 码来确认在第 1 步至第 7 步的工序中没有 DTC 码产生。

4.9 Step 9: Start of Dynamic Test

The process equipment sends out 2 signals in sequences defined in Figure 1 to the IRBC to enable Extended Diagnostic. The operator then drives the vehicle onto the roller tester. Apply the brake while vehicle engine is running to stabilize the vehicle on the roller tester before start of dynamic test.

测试设备依次向 IRBC 发送两个信号（见图 1）来开启 Extended Diagnostic Service。操作员将车辆驶入滚筒试验台，然后踩制动确保在动态测试开始前车辆的平稳。

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4.10 Step 10: Verify Wheel Speed Sensor Input

All the rollers on the tester should be de-coupled from each other. Accelerate the vehicle to maximum test speed, which is defined by OEM. The four wheel speed sensor readings captured by the process equipment shall be <±4% of the true vehicle speed.

首先确认滚筒试验台各滚筒之间是非联动的，然后加速车辆到整车厂指定的最大速度。测试仪器读取到的四个轮速传感器值不应小于实际车速的±4%。

4.11 Step 11: Verify Operation of Base Brake System

Use base brake test procedure defined by OEM to verify Brake Drag, Brake Force, and Brake Balance of the brake system.

根据整车厂定义的基础制动系统检测规范对制动阻力，制动力和制动平衡是否正常。

4.12 Step 12: Verify Operation of Solenoid Valves, Brake Pipe Routing, and Wheel Sensor Wiring

All the rollers on the tester should be de-coupled from each other. The process equipment sends out 2 signals in sequence (as defined in Figure 1) to the IRBC to start Extended Diagnostic Service. The process equipment then sends out InputOutputControlbyIdentifier messages (defined in Section 5.4) to control the On/Off state of pump motor, apply valves, and release valves to complete a series of test sequence.

首先确认滚筒试验台各滚筒之间是非联动的，然后测试设备依次向 IRBC 发送两个信号（见图 1）来开启 Extended Diagnostic Service。测试设备通过向 IRBC 发送 InputOutputControlbyIdentifier 报文（见第 5.4 节）控制 SHA 模块的电机、增压阀和降压阀的开/关来完成一系列测试工序。

Firstly, the operator accelerates the vehicle and the process equipment sends commands to energize valves to build pressure on FL. This is done by energizing pump motor, the cross apply valve, and both isolation and prime valves for the circuit. FL brake force shall increase. Secondly, energize the outlet valve, prime valve, iso valve and the pump motor to release the pressure on FL. All brake forces shall return to none applied state.

首先，操作员加速车辆然后测试设备会发送指令来驱动电磁阀，给左前轮施加制动压力。这个动作是通过控制驱动泵，电机，交叉回路的进液阀还有目标回路的隔离阀和供油阀来实现的。左前轮的制动压力会上升。然后开启泄压阀，供油阀，隔离阀并启动泵电机使左前轮缸减压。最后左前轮的制动压力会回到初始状态（无制动施加）。

Maintain the vehicle speed and repeat the above two steps on FR, RL, and RR wheels.

维持车速，分别对右前轮，左后轮和右后轮重复上述的两个步骤。

After the last valve actuation of all wheels, at the end of the entire test, the pump motor

所有测试完成以后将所有电磁阀复位，待 2 秒钟后关闭电机，测试完成。

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should be actuated for two seconds to discharge the accumulator chamber.

4.13 Step 13: Perform Vacuum Relief Profile

The process equipment sends out 2 signals in sequence (as defined in Figure 1) to the IRBC to start Extended Diagnostic Service. The operator ensures the brake pedal is released and brake switch input is not active. The process equipment then sends out InputOutputControlbyIdentifier messages (defined in Section 5.4) to the IRBC to energize both normally-closed prime valves (open) for 2 seconds with the pump motor off.

测试设备依次向 IRBC 发送两个信号（见图 1）来开启 Extended Diagnostic Service。操作员确认制动踏板已经松开，测试仪器上显示制动踏板状态为松开。测试设备然后向 IRBC 发送 InputOutputControlbyIdentifier 报文（见第 5.4 节定义）开启 SHA 的两个供油阀 2 秒钟，此时确保 SHA 电机处于关闭状态。

4.14 Step 14: Verify DTC Status

Read DTCs stored in the IRBC to ensure no DTCs was set and stored in the IRBC from Step 9 to Step 13.

通过读取 IRBC 中的 DTC 码来确认在第 9 步至第 13 步的工序中没有 DTC 码产生。

4.15 Step 15: End of Test

Clear all DTCs, reset the IRBC, and exit EOL test sequence.

清除所有 DTC 码并重置 IRBC，下线检测流程结束。

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5 Communication 通讯信号

All CAN communication shall use CAN ID 0x731 to send messages to IRBC and CAN ID 0x739 to receive responses.

所有的 CAN 通讯应使用 CAN ID 0x731 来向 IRBC 发送报文，然后使用 CAN ID 0x739 来接收来自 IRBC 的反馈。

5.1 Diagnostic Session

The diagnostic session control service enables diagnostic sessions in the IRBC. A diagnostic session defines applications required for the EOL test process. Messages for diagnostic service are specified as follows:

诊断会话控制服务用于在 IRBC 中开启一个诊断会话。一个诊断会话定义了一系列下线检测所需要的应用。相关的报文信息定义如下：

5.1.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Diagnostic Session Control Request Service ID	M	10
#2	Sub Function = [Diagnostic Session Type] Default Session Extended Diagnostic Session	M	 01 03

5.1.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Diagnostic Session Control Response Service ID	M	50
#2	Sub Function = [Diagnostic Session Type] Default Session Extended Diagnostic Session	M	 01 03
 #3 #4 #5 #6	Session Parameter Record [] =[P2Can_ECU_max (high byte) P2Can_ECU_max (low byte) P2*Can_ECU_max (high byte) P2*Can_ECU_max (low byte)]	 M : : M	 00-FF : : 00-FF

5.1.3 Session Parameter Record Definition

IRBC to Process Equipment

Parameter	Definition	Min	Max
P2Can_ECU_max	Performance requirement for the ECU to start with the response message after the reception of a request message.	0 ms	50 ms
P2*Can_ECU_max	Performance requirement for the ECU to start with a further response message after the transmission of a negative response message.	0 ms	5000 ms

5.1.4 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value

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#1	Negative Response	M	7F
#2	Diagnostic Session Control Request Service ID	M	10
#3	Sub-Function =[Negative Response Trouble Code] Sub-Function Not Supported Incorrect Message Length Or Invalid Format Condition Not Correct	M	 12 13 22

5.2 Tester Present

This service provides indication to IRBC that a diagnostic test tool is connected to the CAN bus and diagnostic sessions and communication have been activated. Tester Present message needs to be sent at a two second periotic rate during Step 7, 9, 12, and 13 of the EOL test process.

此项服务用来提示 IRBC 诊断工具已经连接上 CAN 总线，诊断对话和通讯处于激活状态。Tester Present 报文需要在下线检测的步骤 7，9，12 和 13 中以 2 秒发送一次的周期发送给 IRBC。

5.2.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Tester Present Request Service Id	M	3E
#2	Sub-Function = [Zero Sub-Function] Positive Response Required No Positive Response Required	M	 00 80

5.2.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Tester Present Response Service ID	M	7E
#2	Sub-Function = [Zero Sub-Function] Positive Response Required	M	 00

5.2.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	Tester Present	M	3E
#3	ResponseCode = [Sub-Function Not Supported Incorrect Message Length Or Invalid Format]	M	 12 13

5.3 Input Output Control by Identifier

The InputOutputControlByIdentifier service is primarily used by the IRBC to control the SHA in the brake system. The process Equipment will send IRBC Request message, which contains a

IRBC 通过 InputOutputControlByIdentifier 服务来控制 SHA 内部的阀门和电机。加注设备向 IRBC 发送 Request 报文，此请求报文中包含

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dataIdentifier to reference a group of signals of the IRBC. The IRBC shall send a Positive Response Message if the Request Message can be successfully executed.

The SHA enabled by Input and Output Control commands will remain active for a while (settling time) even after it has been de-activated by Input and Output Control command. When any of the following conditions are met, the IRBC will begin to terminate this service.

- Positively responded to a new diagnostic session request.
- Any Output Control limit is exceeded. See section 5.4.5 Negative Response Message Data Bytes
- The active session is timed out after 5s without receiving any request message (include Tester Present message).

The IRBC will remain in the Input and Output Control until it is determined that the commanded SHA is no longer active. The IRBC will then return control to normal operation. During the SHA settling time, the IRBC is allowed to go back to this service.

有一个引用了一系列 IRBC 用于控制 SHA 信号的 dataIdentifier。如果请求报文成功被执行了，IRBC 会向加注设备发送一个 Positive Response 报文。

即便 Input and Output Control 已经解除了 SHA 的激活状态，先前被此指令启动的 SHA 会依旧保持一段时间（稳定时间）的激活状态。当满足以下任意一个条件时，IRBC 会终止 InputOutputControlByIdentifier 服务：

- IRBC 响应了一个新的诊断请求对话
- 超出任意第 5.4.3 节中定义 Output Control 极值
- 激活的对话没有接收到任何诊断请求报文(包括 Tester Present 报文)超过 5 秒钟

IRBC 会处在 Input and Output Control 服务中，一直到 SHA 不再处于激活的状态。然后 IRBC 会返回到正常工作状态。当还 SHA 处在稳定时间时，IRBC 还可以随时返回到此服务中。

5.3.1 Request Valve / Motor Control Message Definition

Process Equipment to IRBC (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	 31 07
#4	inputOutputControlParameter=[shortTermAdjustment]	M	 03
 #5 #6 #7	ControlOptionRecord #1 = [ControlState#1 ControlState#2 ControlState#3]	 M M M	 00-FF 00-FF 00-FF

Process Equipment to IRBC (returnControlToECU)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB)	M	 31

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	byte #2 (LSB)]		07
#4	inputOutputControlParameter=[returnControlToECU]	M	00

5.3.1.1 Control State Definition

Control State	Bit #	Definition	Units	Range	Command Resolution
#1	7	Rear Right Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	6	Rear Right Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	5	Rear Left Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	4	Rear Left Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	3	Front Right Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	2	Front Right Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	1	Front Left Release Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	0	Front Left Apply Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
#2	7	Primary Prime Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	6	Secondary Prime Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	5	Primary Isolation Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	4	Secondary Isolation Solenoid On-Off Control Command	N/A	N/A	On = 1, Off=0
	3	Not Used.	N/A	N/A	N/A
	2	Not Used.	N/A	N/A	N/A
	1	Pump Motor On/Off Control Command	N/A	N/A	On = 1, Off=0
	0	System Relay Control Command	N/A	N/A	On = 1, Off=0
#3	All	Not Used.	N/A	N/A	N/A

5.3.2 Valve / Motor Control Positive Response Message

IRBC to Process Equipment (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
	dataIdentifier #1 [] = [	M	
#2	byte #1 (MSB)		31
#3	byte #2 (LSB)]		07
#4	inputOutputControlParameter=[shortTermAdjustment]	M	03
	ControlOptionRecord #1 = [		
#5	ControlState#1	M	00-FF
#6	ControlState#2	M	00-FF
#7	ControlState#3]	M	00-FF

IRBC to Process Equipment (returnControlToECU)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	31 07
#4	inputOutputControlParameter=[returnControlToECU]	M	00

5.3.3 Request Lamp Control Message Definition

Process Equipment to IRBC (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	32 00
#4	inputOutputControlParameter=[shortTermAdjustment]	M	03
#4 #5 #6	ControlOptionRecord #1 = [ControlState#1 ControlState#2 ControlState#3]	M M M	00-FF 00-FF 00-FF

Process Equipment to IRBC (returnControlToECU)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Request Sid	M	2F
#2 #3	dataIdentifier #1 [] = [byte #1 (MSB) byte #2 (LSB)]	M	32 00
#4	inputOutputControlParameter=[returnControlToECU]	M	00

5.3.3.1 Control State Definition

Control State	Bit #	Definition	Units	Range	Command Resolution
1#	7	Not Used	N/A	N/A	N/A
	6	ESC OFF Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	5	Brake Warning Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	4	ESC Warning Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	3	ABS Warning Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	2	Airbag Warning Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	1	Seatbelt Request Lamp On-Off Control Command	N/A	N/A	On = 1, Off=0
	0	Not Used	N/A	N/A	N/A

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#2	All	Not Used.	N/A	N/A	N/A
#3	All	Not Used.	N/A	N/A	N/A

5.3.4 Lamp Control Positive Response Message

IRBC to Process Equipment (shortTermAdjustment)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
	dataIdentifier #1 [] = [	M	
#2	byte #1 (MSB)		32
#3	byte #2 (LSB)]		00
#4	inputOutputControlParameter=[shortTermAdjustment]	M	03
	ControlOptionRecord #1 = [		
#5	ControlState#1	M	00-FF
#6	ControlState#2	M	00-FF
#7	ControlState#3]	M	00-FF

IRBC to Process Equipment (returnControlToECU)

Data Byte	Parameter Name	Cvt	Hex Value
#1	inputInputOutputControlByIdentifier Positive Response Sid	S	6F
	dataIdentifier #1 [] = [	M	
#2	byte #1 (MSB)		32
#3	byte #2 (LSB)]		00
#4	inputOutputControlParameter=[returnControlToECU]	M	00

5.3.5 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Cvt	Hex Value
#1	negativeResponse Service Id	S	7F
#2	inputInputOutputControlByIdentifier Request Sid	M	2F
#3	ResponseCode = [	M	
	Incorrect Message Length Or Invalid Format		13
	Conditions Not Correct		22
	Request Out Of Range		31
	Service Not Supported In Active Session		7F
	Motor Or Solenoid Thermal Limit Exceeded		86
	Vehicle Speed Limit Exceeded		88
	System Voltage Out Of Range]		93

5.3.5.1 Response Code Definition

Hex	Condition
13	**incorrectMessageLengthOrInvalidFormat** Wrong length for the specific request
22	**ConditionsNotCorrect**

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	This response code shall be sent if the operating conditions of the ECU are such that it can't perform the required action. The system is currently in initialization or shutdown process. The system is in the process of changing memory.
31	**RequestOutOfRange** Wrong inputOutputLocalIdentifier
7F	**serviceNotSupportedInActiveSession** The service is not enabled in the current diagnostic session.
86	**MotorOrSolenoidThermalLimitExceeded** This code will be returned any time the device exits I/O Control due to its internal protection algorithms. That is, any of the solenoids or motor has been commanded active for an excessive period of time that may begin to damage the device. Please note that this response can be send unsolicited.
88	**VehicleSpeedLimitExceeded** This code will be returned any time the device exits I/O Control due to an unsafe condition. That is, the vehicle speed 5 KPH, while the device's is not currently in SafetySystemDiagnosticSession. Please note that this response can be send unsolicited.
93	**SystemVoltageOutOfRange** This code will be returned any time the device exits I/O Control due to the ignition voltage not been sufficient Please note that this response can be send unsolicited.

5.4 Clear DTC

5.4.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Clear Diagnostic Information Request Service ID	M	14
#2- #4	Clear All DTCs	M	FFFFFF

5.4.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Clear Diagnostic Information Positive Response	M	54

5.4.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	Clear Diagnostic Information Positive Response	M	14
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

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5.5 Read DTC

5.5.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DTC Information Request Service ID	M	19
#2	Sub function =[Report Number Of DTC By Status Mask Report DTC By Status Mask]	M	 01 02
#3	DTC Status Mask	M	8B

5.5.2 Positive Response Message for Sub Function 01

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DTC Information Response Service ID	M	59
#2	Report DTC By Status Mask	M	01
#3	DTC Status Availability Mask	M	8B
#4	DTC Format Identifier	M	00
 #5 #6	DTC Count =[DTC Count High Byte DTC Count Low Byte	 M M	 00-FF 00-FF

5.5.3 Positive Response Message for Sub Function 02

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DTC Information Response Service ID	M	59
#2	Report DTC By Status Mask	M	02
#3	DTC Status Availability	M	8B
	Data Record[]=[		
#4	DTC no. 1 high Byte	C1	00-FF
#5	DTC no. 1 middle Byte	C1	00-FF
#6	DTC no. 1 low Byte	C1	00-FF
#7	Status of DTC no. 1	C1	00-FF
#8	DTC no.2 high Byte	C2	00-FF
#9	DTC no. 2 middle Byte	C2	00-FF
#10	DTC no. 2 low Byte	C2	00-FF
#11	Status of DTC no. 2	C2	00-FF
:	:	:	:
n-3	DTC no. m high Byte	C2	00-FF
n-2	DTC no. m middle Byte	C2	00-FF
n-1	DTC no. m low Byte	C2	00-FF
n	Status of DTC no. m]	C2	00-FF

Note: C1: This parameter is only present if solely one DTC information is available to be reported.
C2: This parameter is only present if more than one DTC information is available to be reported.

5.5.4 Negative Response Message

IRBC to Process Equipment

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Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	Clear Diagnostic Information Positive Response	M	19
#3	ResponseCode = [Sub-Function Not Supported Incorrect Message Length Or Invalid Format Condition Not Correct]	M	 12 13 22

5.6 Read IRBC Part Number

5.6.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Request Service ID	M	22
#2 - #3	Part Number DID	M	F1CB

5.6.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Response Service ID	M	62
#2 - #3	Part Number DID	M	F1CB
#4 : #7	Part Number (Hex)	M	00-FF : 00-FF

5.6.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	DID Request Service	M	22
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

5.7 Read SHA Part Number

5.7.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Request Service ID	M	22
#2 - #3	Part Number DID	M	F113

5.7.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Response Service ID	M	62

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#2 - #3	Part Number DID	M	F113
#4 : #14	Part Number (ASCII)	M	00-FF : 00-FF

5.7.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	DID Request Service	M	22
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

5.8 IRBC Reset

5.8.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Reset Service Id	M	11
#2	Sub-Function = [Zero Sub-Function] Hard Reset	M	 01

5.8.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Reset Service ID	M	51
#2	Sub-Function = [Zero Sub-Function]		01

5.8.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	Reset Service ID	M	11
#3	ResponseCode = [Sub-Function Not Supported Incorrect Message Length Or Invalid Format Conditions Not Correct]	M	 12 13 22

5.9 Read Master Cylinder Pressure

5.9.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Request Service ID	M	22

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#2 - #3	System Sensor DID	M	2D05

5.9.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Response Service ID	M	62
#2 - #3	System Sensor DID	M	2D05
#4	Master Cylinder Pressure (HEX) MSB	M	00-FF
#5	Master Cylinder Pressure (HEX) LSB	M	00-FF
#6 : #13	Other system sensor information	M	00-FF : 00-FF

Note: Master Cylinder Pressure unit: bar, scaling: 0.0153, offset: 0, physical limits: -42.5 ~ 42.5

5.9.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	DID Request Service	M	22
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

5.10 Read Wheel Speed

5.10.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Request Service ID	M	22
#2 - #3	SpeedInformation DID	M	2D00

5.10.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Response Service ID	M	62
#2 - #3	SpeedInformation DID	M	2D00
#4 - #5	Other information	M	0000-FFFF
#6	WheelSpeedFL (HEX) MSB	M	00-FF
#7	WheelSpeedFL (HEX) LSB	M	00-FF
#8	WheelSpeedFR (HEX) MSB	M	00-FF
#9	WheelSpeedFR (HEX) LSB	M	00-FF
#10	WheelSpeedRL (HEX) MSB	M	00-FF
#11	WheelSpeedRL (HEX) LSB	M	00-FF
#12	WheelSpeedRR (HEX) MSB	M	00-FF
#13	WheelSpeedRR (HEX) LSB	M	00-FF

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Note: Wheel Speed unit: km/h, scaling: 0.05625, Hex limit: 0x0000 ~ 0x120E, physical limit: 0 ~ 260
If Invaild, send out 0x7FFF

5.10.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	DID Request Service	M	22
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

5.11 Read Steering Angle

5.11.1 Request Message Definition

Process Equipment to IRBC

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Request Service ID	M	22
#2 - #3	System Sensor DID	M	2D05

5.11.2 Positive Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Read DID Response Service ID	M	62
#2 - #3	System Sensor DID	M	2D05
#4 - #5	Other system sensor information	M	00-FF
#6	Steering Angle (HEX) MSB	M	00-FF
#7	Steering Angle (HEX) LSB	M	00-FF
#8 : #13	Other system sensor information	M : M	00-FF : 00-FF

Note: Steering Angle Sensor unit: deg, scaling: 0.1, offset: 0, physical limits: -1440 ~ 1440

5.11.3 Negative Response Message

IRBC to Process Equipment

Data Byte	Parameter Name	Message Usage	Hex Value
#1	Negative Response	M	7F
#2	DID Request Service	M	22
#3	ResponseCode = [Incorrect Message Length Or Invalid Format Condition Not Correct Request out of Range]	M	 13 22 31

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